

VIRTUAL INTERNSHIP · JUNIOR WEB DEVELOPER

PashuMitra

A Ward-Level Stray Animal Grievance & Care Coordination Platform

A 4-week project covering requirements planning, responsive UI design, QA & testing strategy, and a pre-launch performance/accessibility/security audit — for an e-governance digital service.

Bhoomi · B.Tech CSE, IILM University, Greater Noida

Four Weeks, One Continuous Project

01 Planning & Requirements

Stakeholder personas, functional/non-functional requirements, risk analysis

02 Responsive Web Design

Site map, high-fidelity mockups across breakpoints, accessibility-first UI

03 QA & Testing Strategy

Test pyramid, 7 detailed test cases, CI/CD pipeline, risk heatmap

04 Performance, Accessibility & Security Audit

Core Web Vitals, WCAG 2.1 AA compliance, OWASP-based findings

PLUS

A Working Prototype

Beyond the four weekly reports, a functional front-end build simulates the citizen portal end-to-end — real form validation, live case tracking, and a working dashboard.

Stray Animal Grievances Don't Fit a Generic Complaint Portal

- **A live, moving subject**

Unlike a pothole or a garbage pile, the case involves an animal that may need to be located again — location and photo evidence matter more.

- **A defined legal process**

The Animal Birth Control (ABC) Rules, 2023 already set out capture → sterilise → vaccinate → release/shelter as the required workflow.

- **Multiple disconnected actors**

Citizens, ward officers, AWBI-recognised NGOs, and vets rarely share one system — cases get lost between a helpline and a spreadsheet.

- **Real public-health urgency**

Bite and rabies-related complaints need faster, trackable response than a routine civic complaint.

Planning & Requirements Analysis

- Benchmarked existing initiatives — Panvel's ABC Helpline, MCD's feeding-point registry, DigiLocker/UMANG identity patterns
- Defined 4 stakeholder personas: resident reporter, community feeder, veterinary officer, ward officer
- 12 functional requirements (FR-01 to FR-12) prioritised Must/Should/Could
- 9 non-functional requirements spanning availability, security, accessibility, and offline resilience
- Full risk analysis with likelihood/impact ratings and mitigation strategies

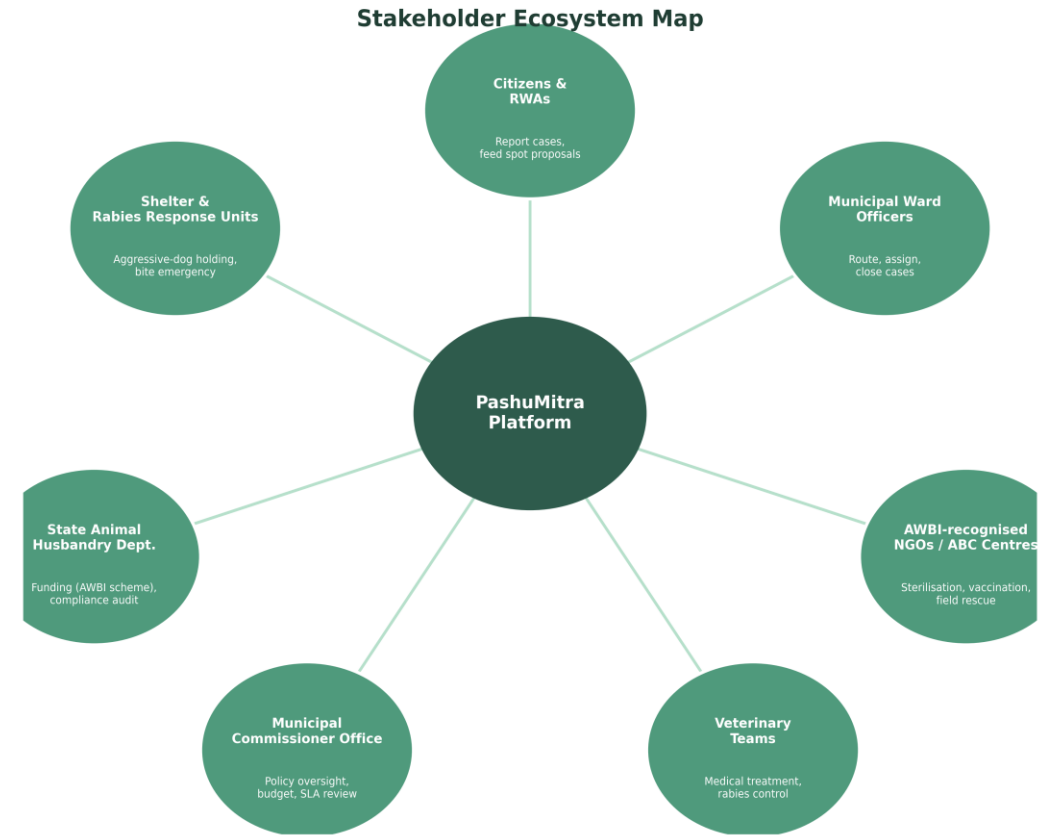


Figure — Stakeholder ecosystem around the platform

Responsive Web Prototype Design

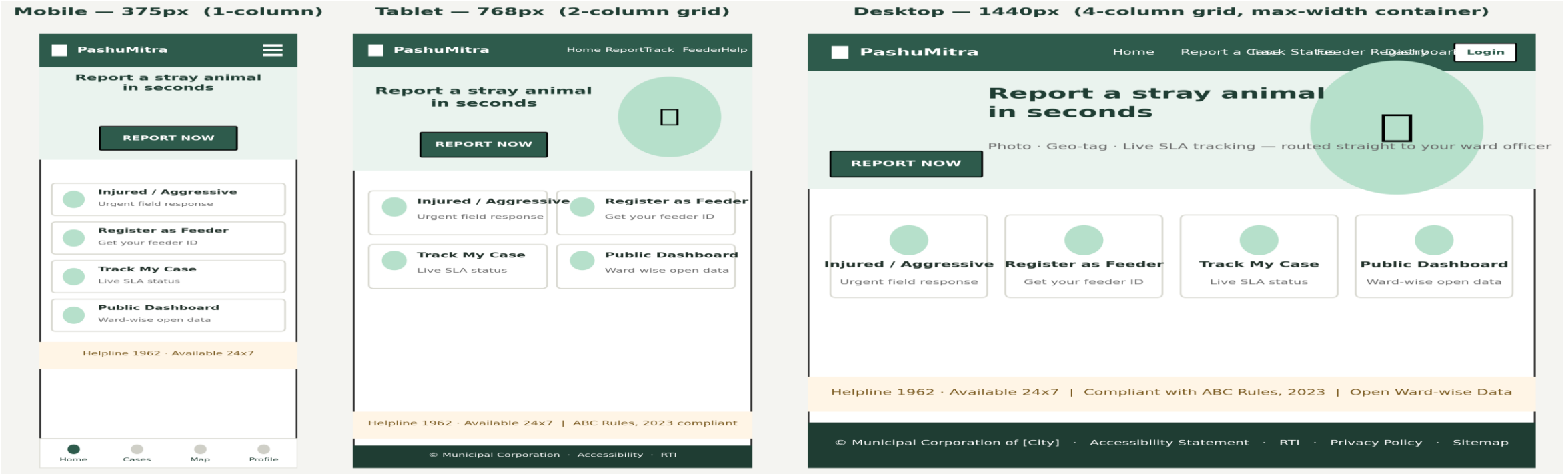
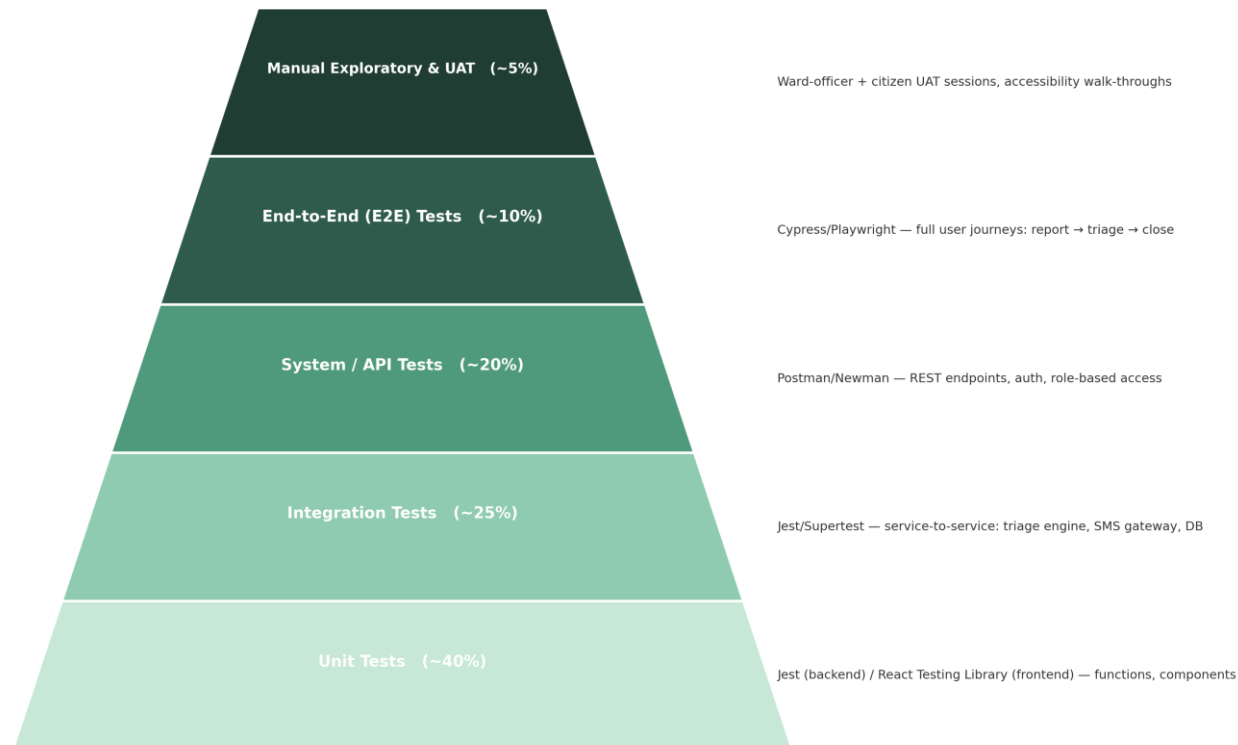


Figure — Homepage layout at 375px (mobile), 768px (tablet), and 1440px (desktop)

- Mobile-first, fluid grid
- WCAG 2.1 AA target
- One-tap photo-first report flow
- GIGW-aligned trust cues

QA & Testing Strategy

PashuMitra Test Pyramid — Automation-First Strategy



7

Detailed test cases

8

Test types covered

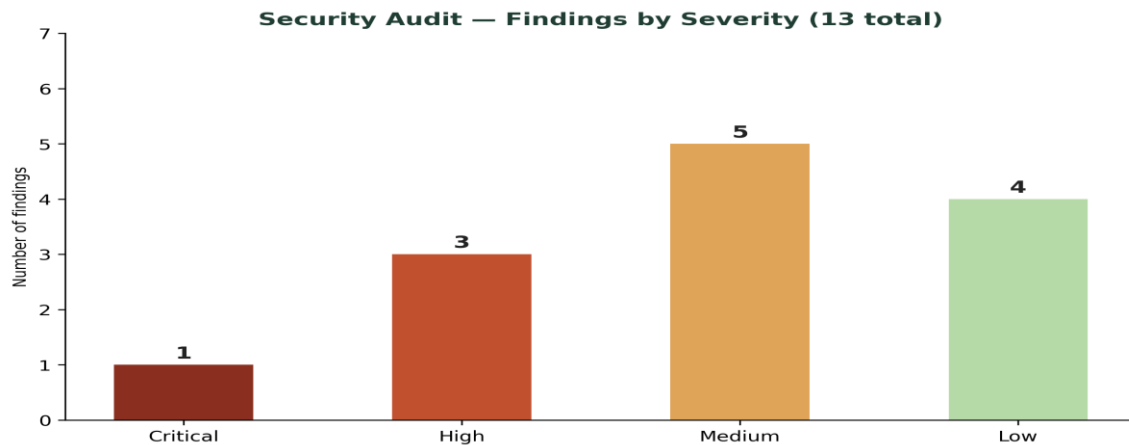
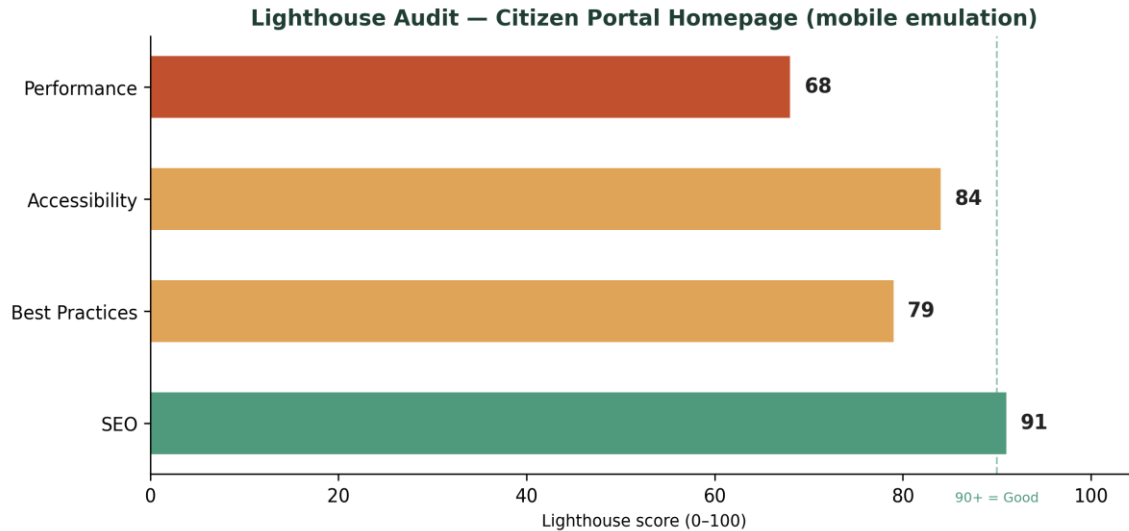
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Risks mapped & mitigated

4

Defect severity tiers

Performance, Accessibility & Security Audit



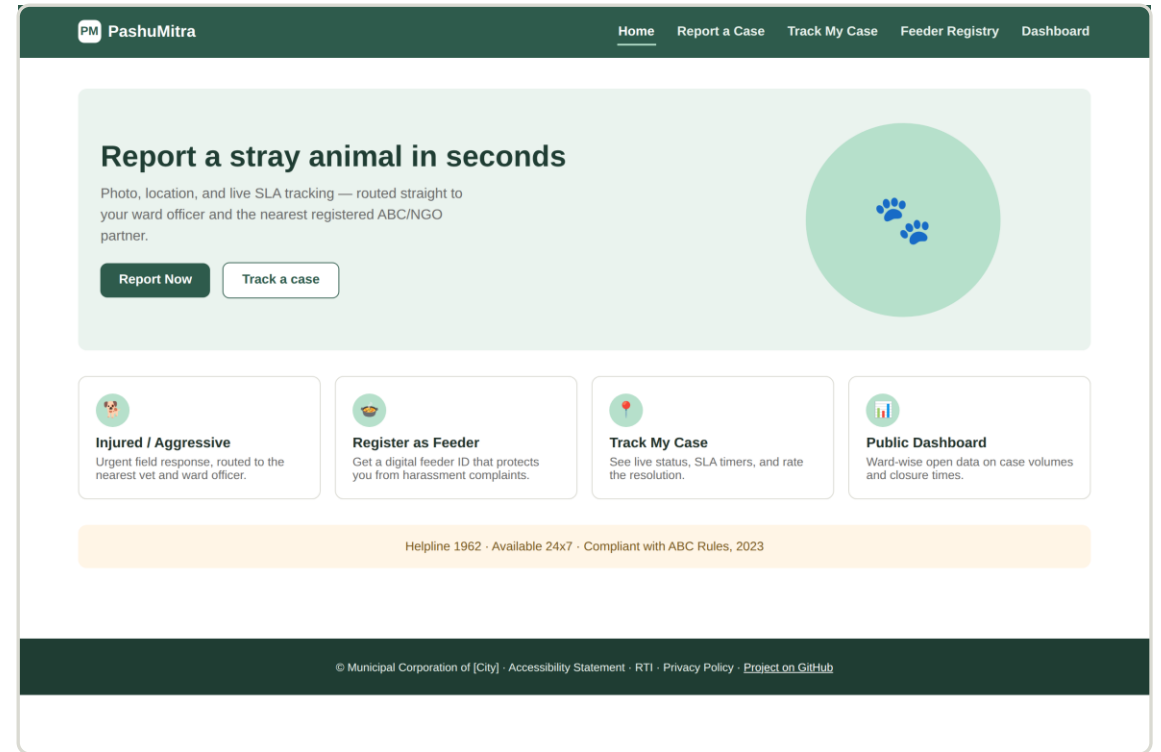
Top findings

- Critical: broken access control let a citizen token reach an admin export endpoint
- High: stored XSS risk in the case-description field, unsanitised
- High: PII (phone numbers) exposed unmasked to NGO field app
- Performance: LCP 3.1s on 3G vs. 2.5s target — hero image needs optimisation
- Accessibility: 65% Operable-principle compliance — keyboard nav gaps on upload control

6 P0 issues fixed before pilot launch · 6 P1 in first sprint

A Working Front-End Prototype

- Real client-side validation — blocks submission until name, phone, category, and photo are filled
- Case submission generates a live case ID and feeds directly into Track My Case
- Status stepper with an "Advance status" control simulating the real backend workflow
- Feeder registry issues a working digital ID on submit
- Dashboard stats and category chart update live from session data
- Fully responsive — tested at 375px mobile and 1280px desktop, keyboard-navigable throughout



Live screenshot — citizen portal homepage

Technology Stack



FRONTEND

React Native (mobile) + React.js (web)

Mobile-first, shared logic across citizen app and ward-officer console



BACKEND

Node.js + Express

Lightweight REST services for the case-engine API



DATABASE

PostgreSQL + PostGIS

Relational integrity plus geo-fenced ward routing



TESTING

Jest, Cypress, k6, OWASP ZAP, axe-core

Covers unit through security and accessibility layers



HOSTING

MeitY GI Cloud ("MeghRaj")

Empanelled cloud, standard for Indian e-governance deployments



AUTH

Aadhaar-linked OTP + SSO-ready staff login

Consistent with DigiLocker/UMANG identity patterns

ACROSS ALL FOUR WEEKS

Project at a Glance

4

Weeks of continuous
requirements → design → QA → audit

12

Functional requirements
defined and traced through every phase

7

Detailed test cases
spanning functional, load, and security

13

Audit findings
prioritised P0/P1/P2 for pilot launch

Every screen and test traces back to a specific requirement — Week 1 defined what to build, Week 2 defined how it should look, Week 3 defined how to verify it, and Week 4 checked the actual build against measurable benchmarks.

THANK YOU

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Full reports, diagrams, and the working prototype are available on GitHub.